

Lab Experiments

Set(21-25)

Date:12-08-26

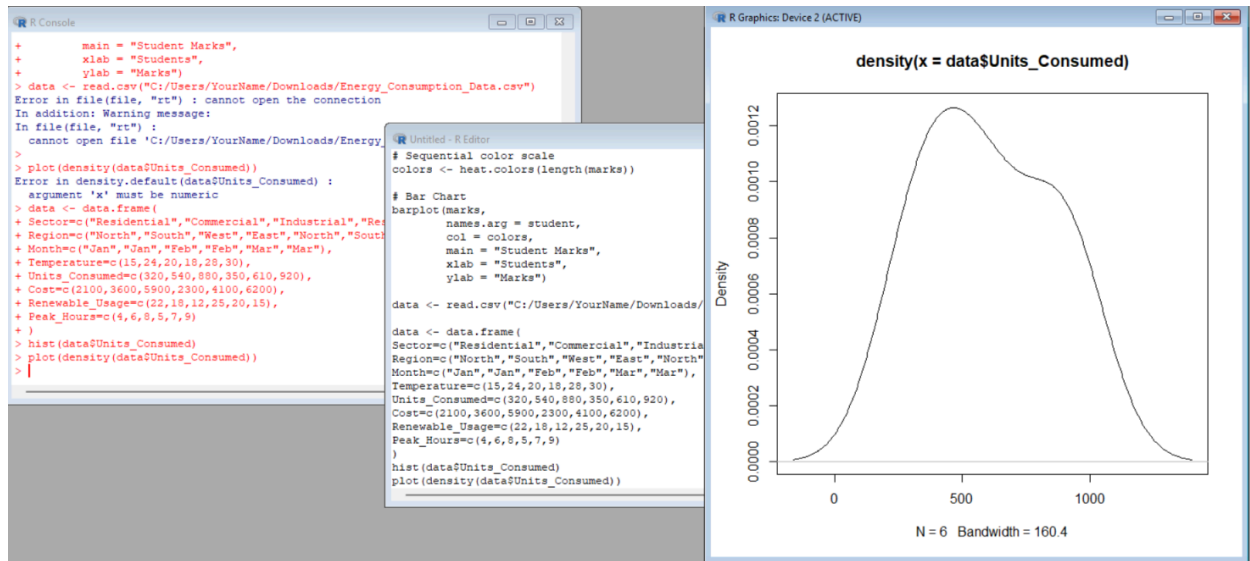
Set 21.Dataset: Energy_Consumption_Data.csv

1. Import the dataset. Plot Histogram and Density Plot

Code:

```
# Import dataset
data <- read.csv("Energy_Consumption_Data.csv")
# Histogram
hist(data$Units_Consumed,
     main="Histogram of Units Consumed",
     xlab="Units Consumed (kWh)",
     col="lightblue")
# Density Plot
plot(density(data$Units_Consumed),
     main="Density Plot of Units Consumed",
     xlab="Units Consumed")
```

Output:



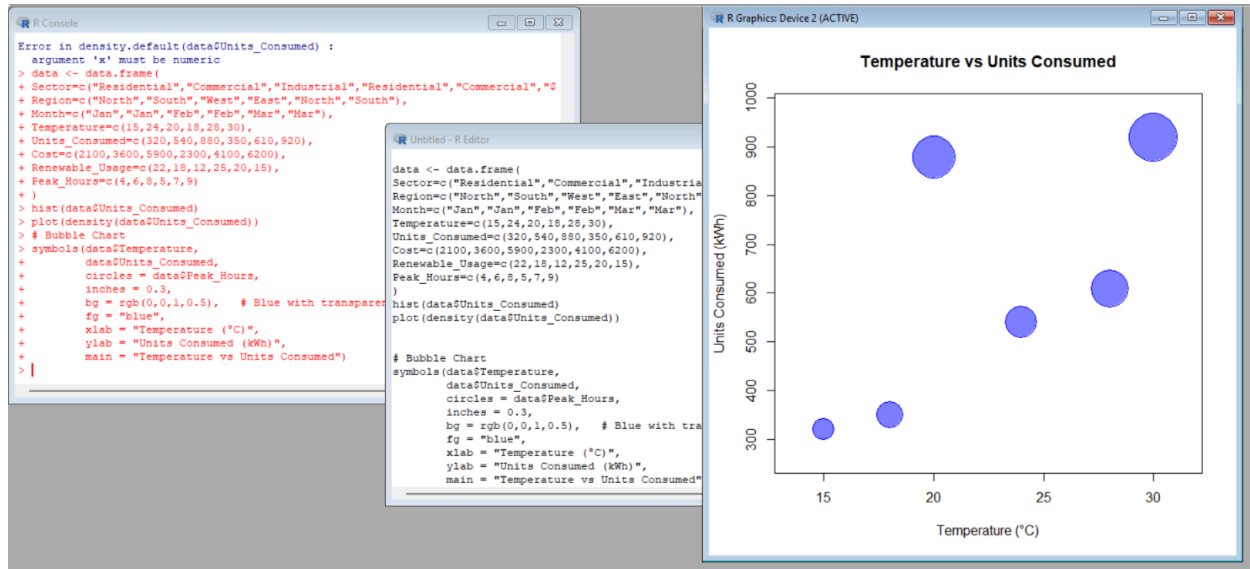
2. Scatter Plot (Bubble Chart) in R

Code:

```
# Bubble Chart
symbols(data$Temperature,
       data$Units_Consumed,
       circles = data$Peak_Hours,
       inches = 0.3,
       bg = rgb(0,0,1,0.5), # Blue with transparency
```

```
fg = "blue",
xlab = "Temperature (°C)",
ylab = "Units Consumed (kWh)",
main = "Temperature vs Units Consumed")
```

Output:

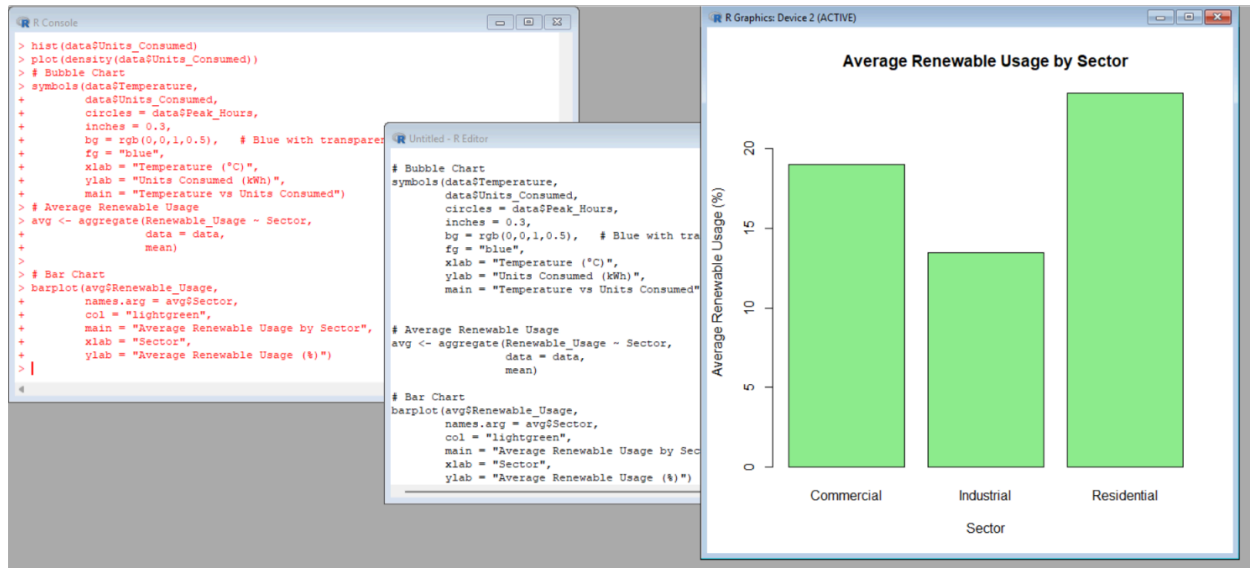


3. Average Renewable Usage by Sector

Code:

```
# Average Renewable Usage
avg <- aggregate(Renewable_Usage ~ Sector,
  data = data,
  mean)
# Bar Chart
barplot(avg$Renewable_Usage,
  names.arg = avg$Sector,
  col = "lightgreen",
  main = "Average Renewable Usage by Sector",
  xlab = "Sector",
  ylab = "Average Renewable Usage (%)")
```

Output:



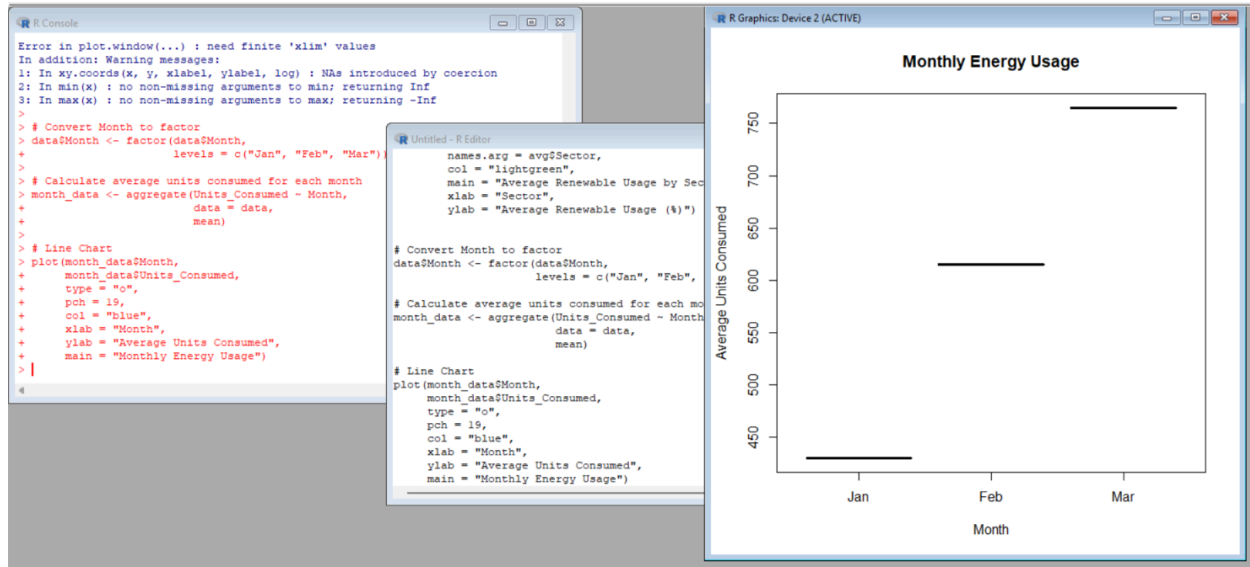
Question 4 (By Using R Program)

1. Line Chart for Monthly Energy Usage

Code:

```
# Convert Month to factor
data$Month <- factor(data$Month,
                     levels = c("Jan", "Feb", "Mar"))
# Calculate average units consumed for each month
month_data <- aggregate(Units_Consumed ~ Month,
                        data = data,
                        mean)
# Line Chart
plot(month_data$Month,
     month_data$Units_Consumed,
     type = "o",
     pch = 19,
     col = "blue",
     xlab = "Month",
     ylab = "Average Units Consumed",
     main = "Monthly Energy Usage")
```

Output:

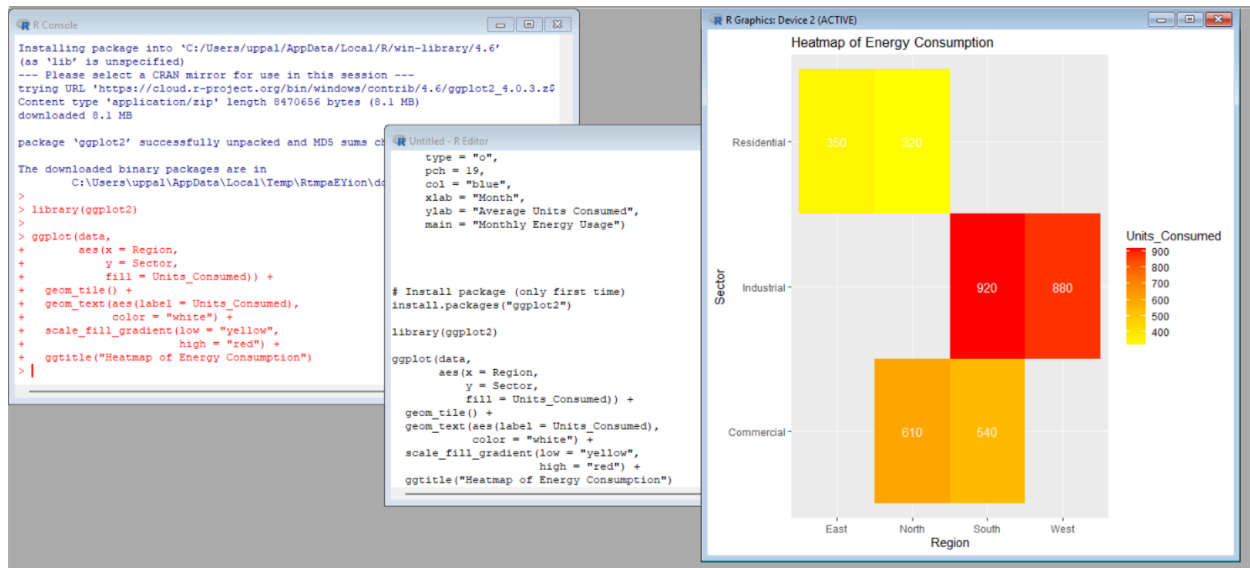


2. Heatmap in R

Code:

```
# Install package (only first time)
install.packages("ggplot2")
library(ggplot2)
ggplot(data,
  aes(x = Region,
    y = Sector,
    fill = Units_Consumed)) +
  geom_tile() +
  geom_text(aes(label = Units_Consumed),
    color = "white") +
  scale_fill_gradient(low = "yellow",
    high = "red") +
  ggtitle("Heatmap of Energy Consumption")
```

Output:



Set 22: Time Series Analysis

1. Time Series Line Chart (R)

Code:

Dataset

```
month <- c("January", "February", "March", "April", "May")
```

```
sales <- c(15000, 18000, 22000, 20000, 23000)
```

Line Chart

```
plot(1:5, sales,
```

```
  type="o",
```

```
  pch=19,
```

```
  col="blue",
```

```
  xaxt="n",
```

```
  xlab="Month",
```

```
  ylab="Sales ($)",
```

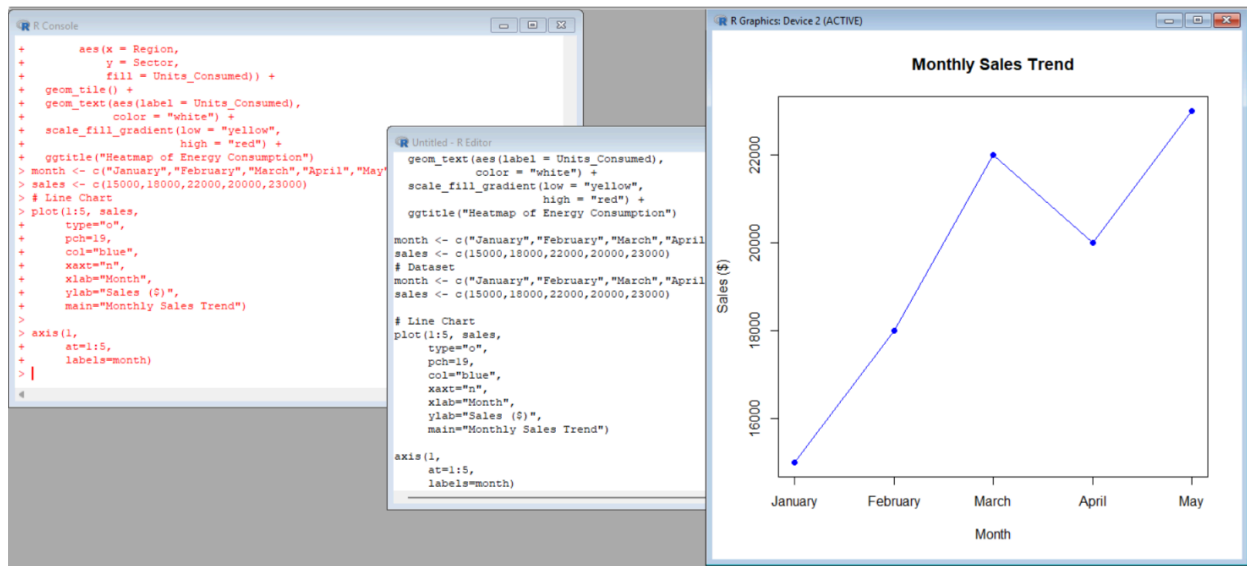
```
  main="Monthly Sales Trend")
```

```
axis(1,
```

```
  at=1:5,
```

```
  labels=month)
```

Output:



2. Scatter Plot (Advertising Budget vs Sales)

Code:

```
# Advertising Budget
```

```
advertising <- c(3000,3500,4000,3800,4500)
```

```
# Scatter Plot
```

```
plot(advertising,
```

```
     sales,
```

```
     pch=19,
```

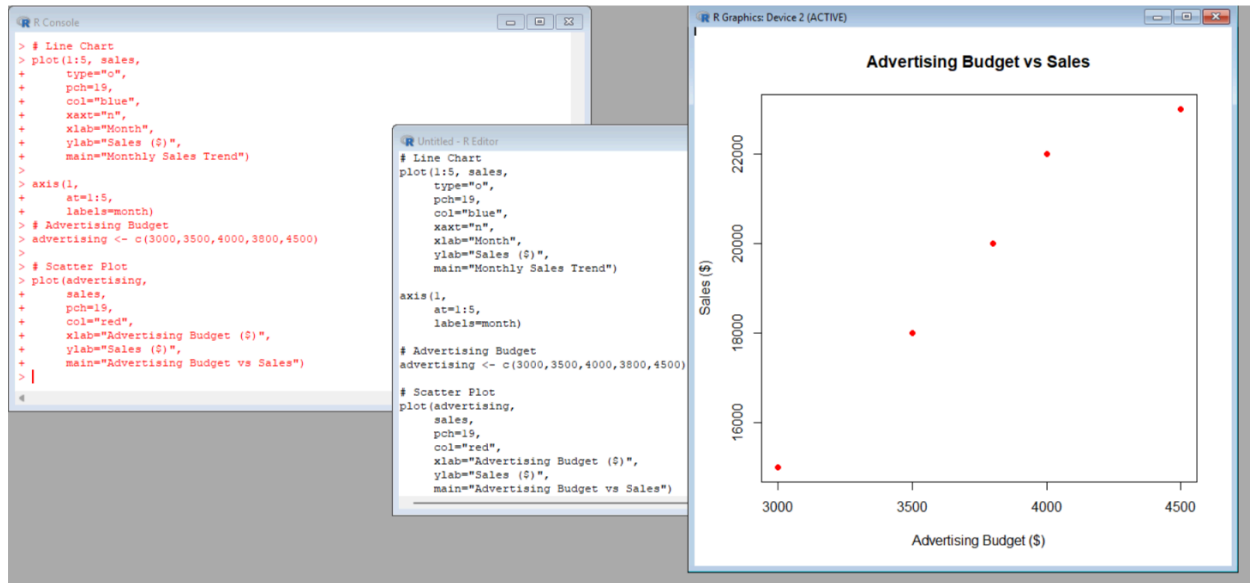
```
     col="red",
```

```
     xlab="Advertising Budget ($)",
```

```
     ylab="Sales ($)",
```

```
     main="Advertising Budget vs Sales")
```

Output:



3. By using R Program

Code:

```
# Dataset
month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)

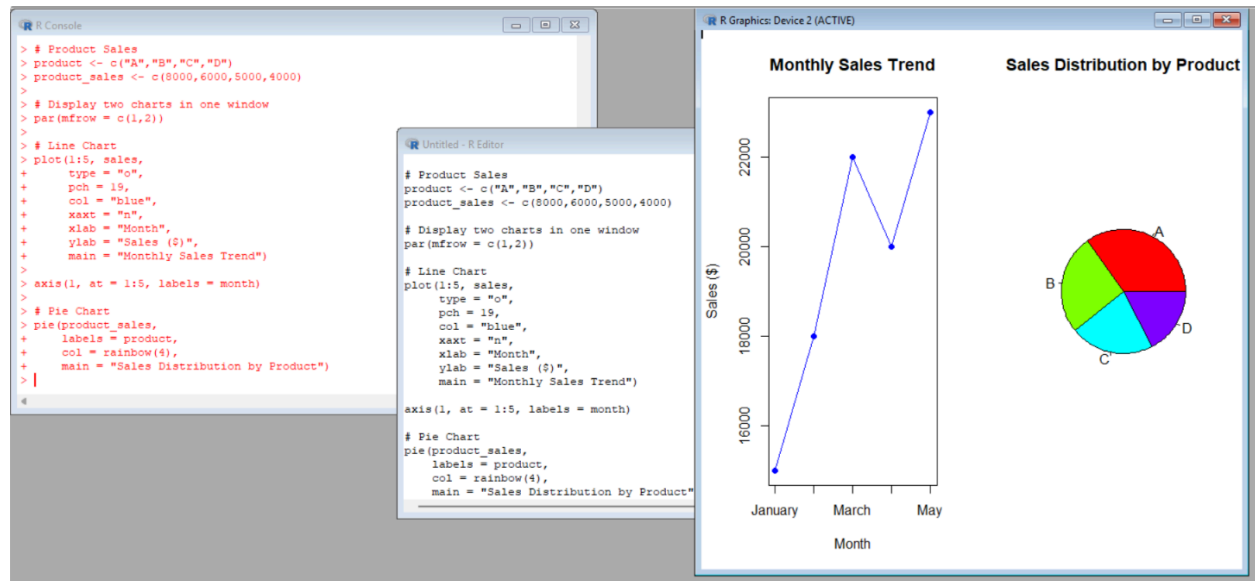
# Product Sales
product <- c("A", "B", "C", "D")
product_sales <- c(8000, 6000, 5000, 4000)

# Display two charts in one window
par(mfrow = c(1, 2))

# Line Chart
plot(1:5, sales,
     type = "o",
     pch = 19,
     col = "blue",
     yaxt = "n",
     xlab = "Month",
     ylab = "Sales ($)",
     main = "Monthly Sales Trend")
axis(1, at = 1:5, labels = month)

# Pie Chart
pie(product_sales,
     labels = product,
     col = rainbow(4),
     main = "Sales Distribution by Product")
```

Output:



4. Autocorrelation Plot (R)

Code:

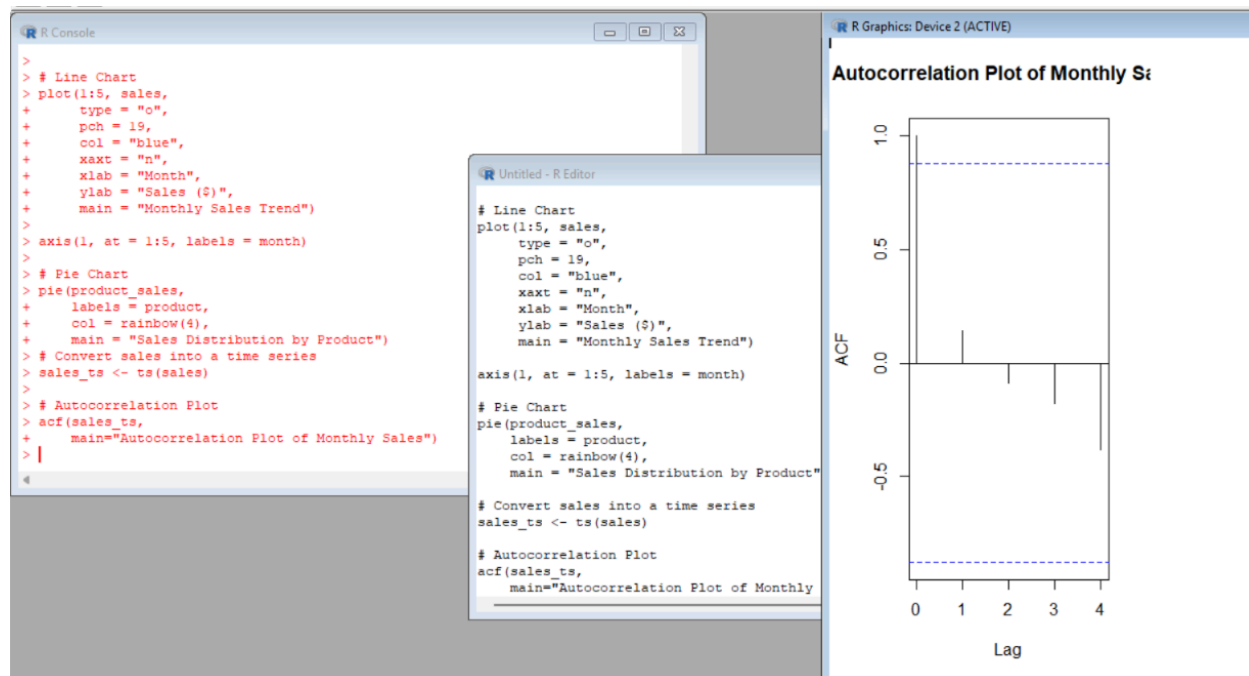
```
# Convert sales into a time series
```

```
sales_ts <- ts(sales)
```

```
# Autocorrelation Plot
```

```
acf(sales_ts,
    main="Autocorrelation Plot of Monthly Sales")
```

Output:



Set 23: Employee Performance Analysis

1. Bar Chart (Department Distribution)

Code:

```
# Count employees in each department
dept <- table(data$department)
# Bar Chart
barplot(dept,
        col="skyblue",
        main="Employee Distribution by Department",
        xlab="Department",
        ylab="Number of Employees")
```

Output:



2. Line Chart (Performance Trend)

Code:

Line Chart

```

plot(data$years,
     data$score,
     type="o",
     pch=19,
     col="blue",
     xlab="Years of Service",
     ylab="Performance Score",
     main="Employee Performance Trend")

```

Output:

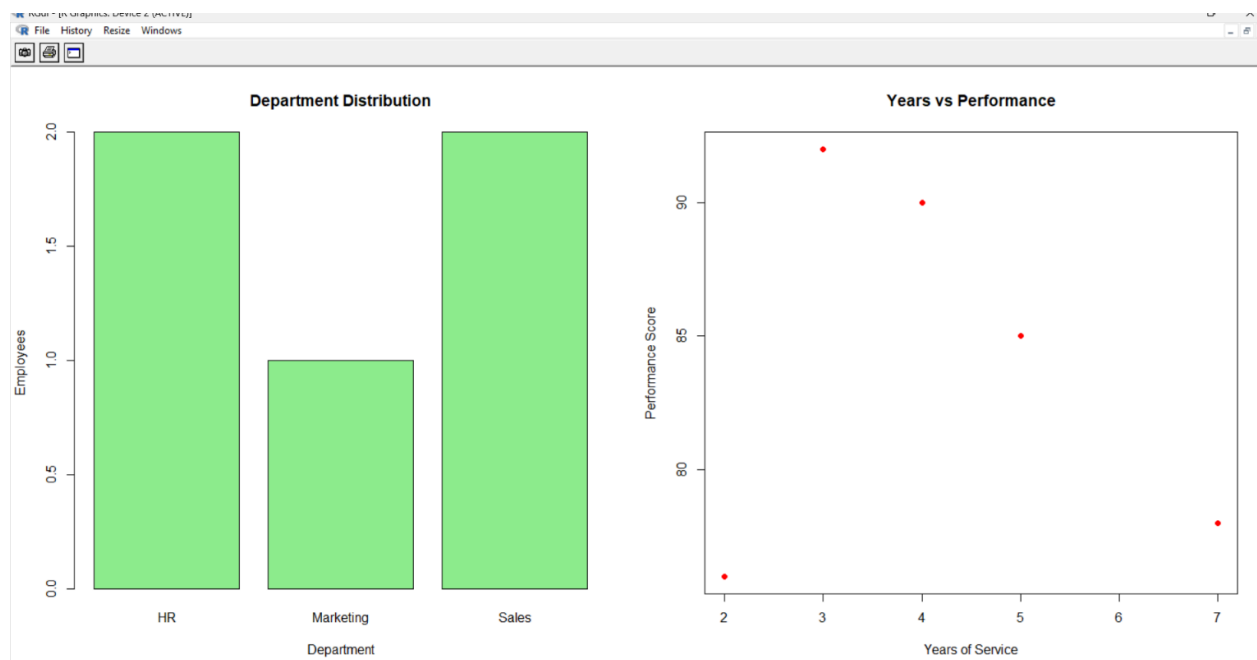


3. Dashboard (R Version)

Code:

```
# Display two charts together
par(mfrow=c(1,2))
# Bar Chart
barplot(table(data$department),
        col="lightgreen",
        main="Department Distribution",
        xlab="Department",
        ylab="Employees")
# Scatter Plot
plot(data$years,
     data$score,
     pch=19,
     col="red",
     xlab="Years of Service",
     ylab="Performance Score",
     main="Years vs Performance")
```

Output:



4. Performance Table (R)

Code:

```
# Display Table
performance_table <- data.frame(
  Employee_ID = employee,
  Department = department,
  Years_of_Service = years,
  Performance_Score = score
)
```

```
print(performance_table)
```

Output:

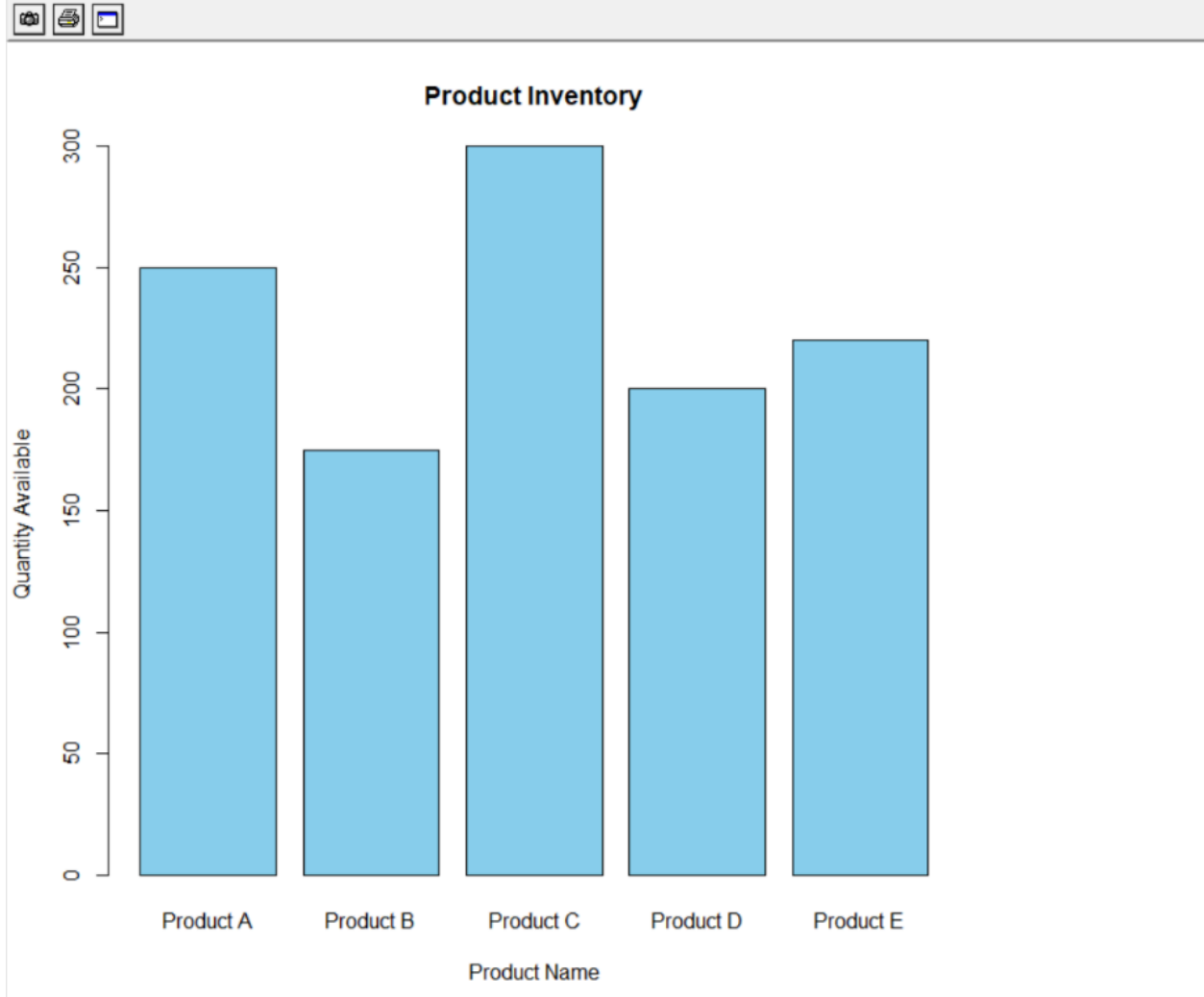
```
>
> print(performance_table)
  Employee_ID Department Years_of_Service Performance_Score
1           1      Sales                5                85
2           2        HR                 3                92
3           3  Marketing                7                78
4           4      Sales                4                90
5           5        HR                 2                76
> |
```

Set 24: Product Inventory Management

1. Bar Chart

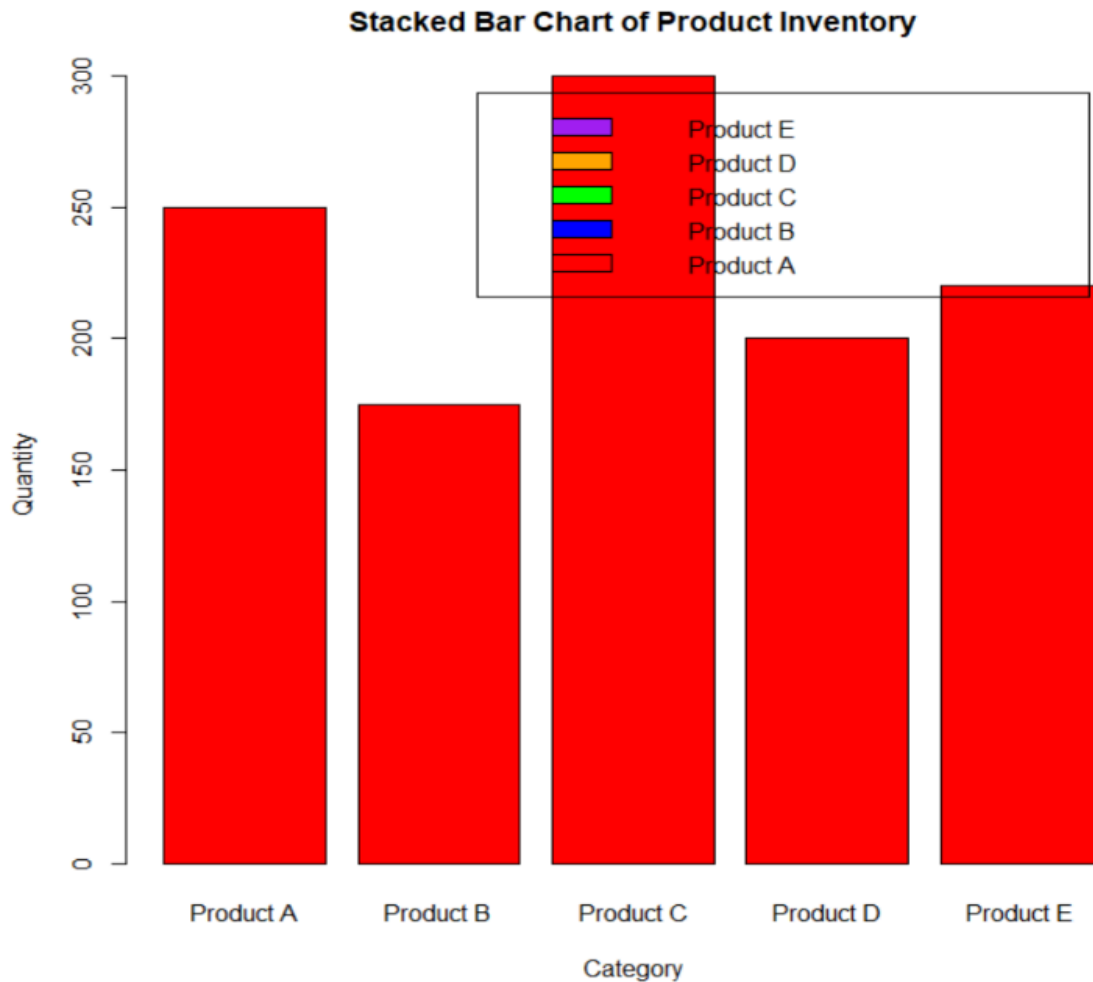
Code:

```
# Dataset
product <- c("Product A","Product B","Product C","Product D","Product E")
quantity <- c(250,175,300,200,220)
data <- data.frame(product, quantity)
# Bar Chart
barplot(data$quantity,
        names.arg=data$product,
        col="skyblue",
        main="Product Inventory",
        xlab="Product Name",
        ylab="Quantity Available")
```

Output:**2. Stacked Bar Chart**

Code:

```
# Product Categories
category <- c("Electronics", "Electronics",
             "Furniture", "Furniture",
             "Accessories")
# Matrix for stacked bar chart
inventory <- matrix(quantity,
                    nrow=1,
                    dimnames=list("Quantity", product))
barplot(inventory,
        col=c("red", "blue", "green", "orange", "purple"),
        main="Stacked Bar Chart of Product Inventory",
        xlab="Category",
        ylab="Quantity",
        legend.text=product)
```

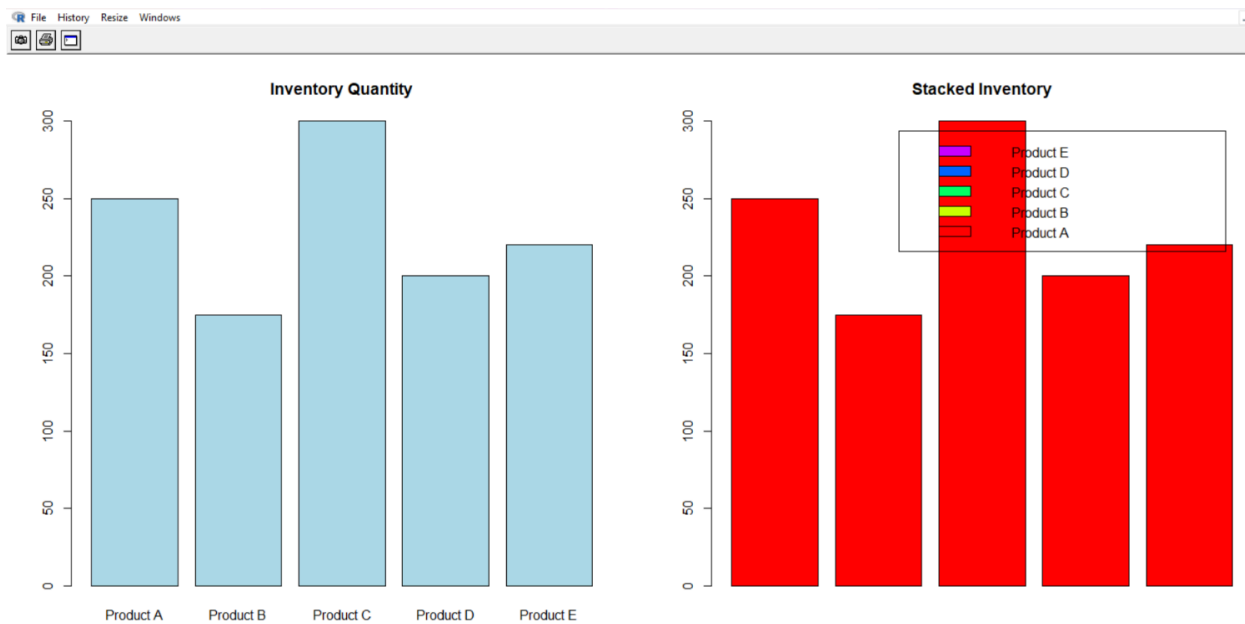
Output:

3. Dashboard (R Version)

Code:

```
# Display two charts together
par(mfrow=c(1,2))
# Bar Chart
barplot(quantity,
        names.arg=product,
        col="lightblue",
        main="Inventory Quantity")
# Stacked Bar Chart
inventory <- matrix(quantity,nrow=1)
barplot(inventory,
        col=rainbow(5),
        legend.text=product,
        main="Stacked Inventory")
```

Output:



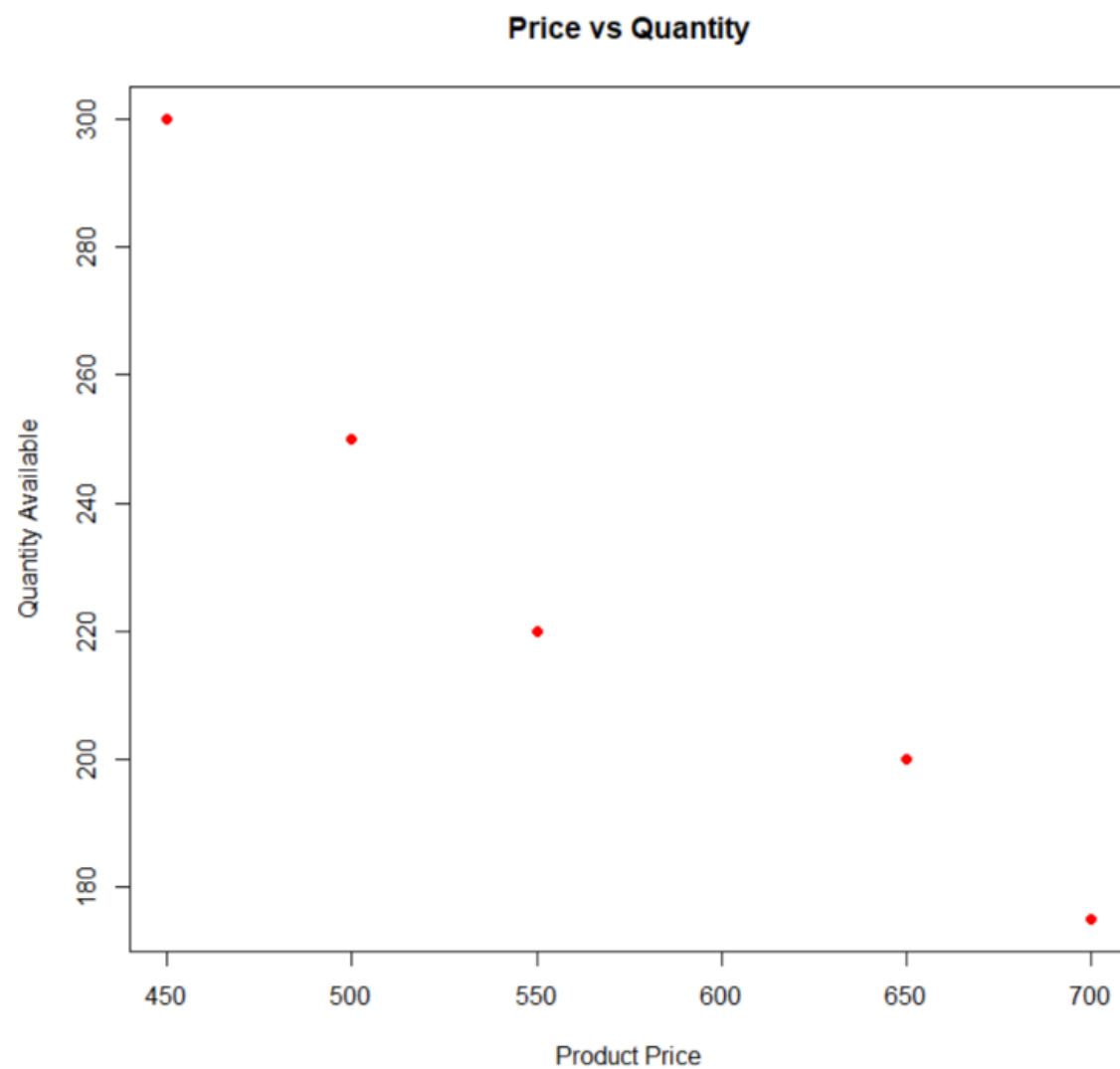
4. Scatter Plot (Price vs Quantity)

Code:

```
# Product Prices
```

```
price <- c(500,700,450,650,550)
# Scatter Plot
plot(price,
      quantity,
      pch=19,
      col="red",
      xlab="Product Price",
      ylab="Quantity Available",
      main="Price vs Quantity")
```

Output:



Set 25: Website Traffic Analysis

1. R Program – Line Chart for Website Traffic Trend

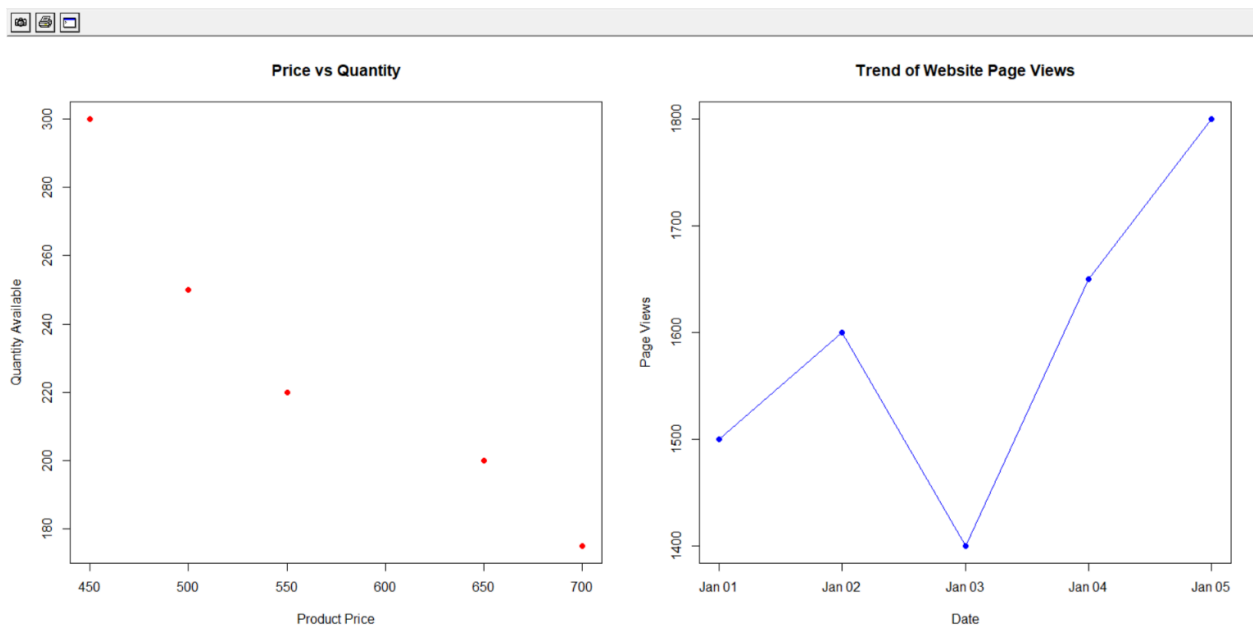
Code:

```
# Dataset
date <- as.Date(c("2023-01-01",
                  "2023-01-02",
                  "2023-01-03",
                  "2023-01-04",
                  "2023-01-05"))

page_views <- c(1500,1600,1400,1650,1800)

# Line Chart
plot(date,
      page_views,
      type="o",
      pch=19,
      col="blue",
      xlab="Date",
      ylab="Page Views",
      main="Trend of Website Page Views")
```

Output:



2. R – Bar Chart (Top N Click-through Rates)

Code:

```
# Dataset
date <- c("2023-01-01","2023-01-02","2023-01-03",
          "2023-01-04","2023-01-05")

page_views <- c(1500,1600,1400,1650,1800)
ctr <- c(2.3,2.7,2.0,2.4,2.6)

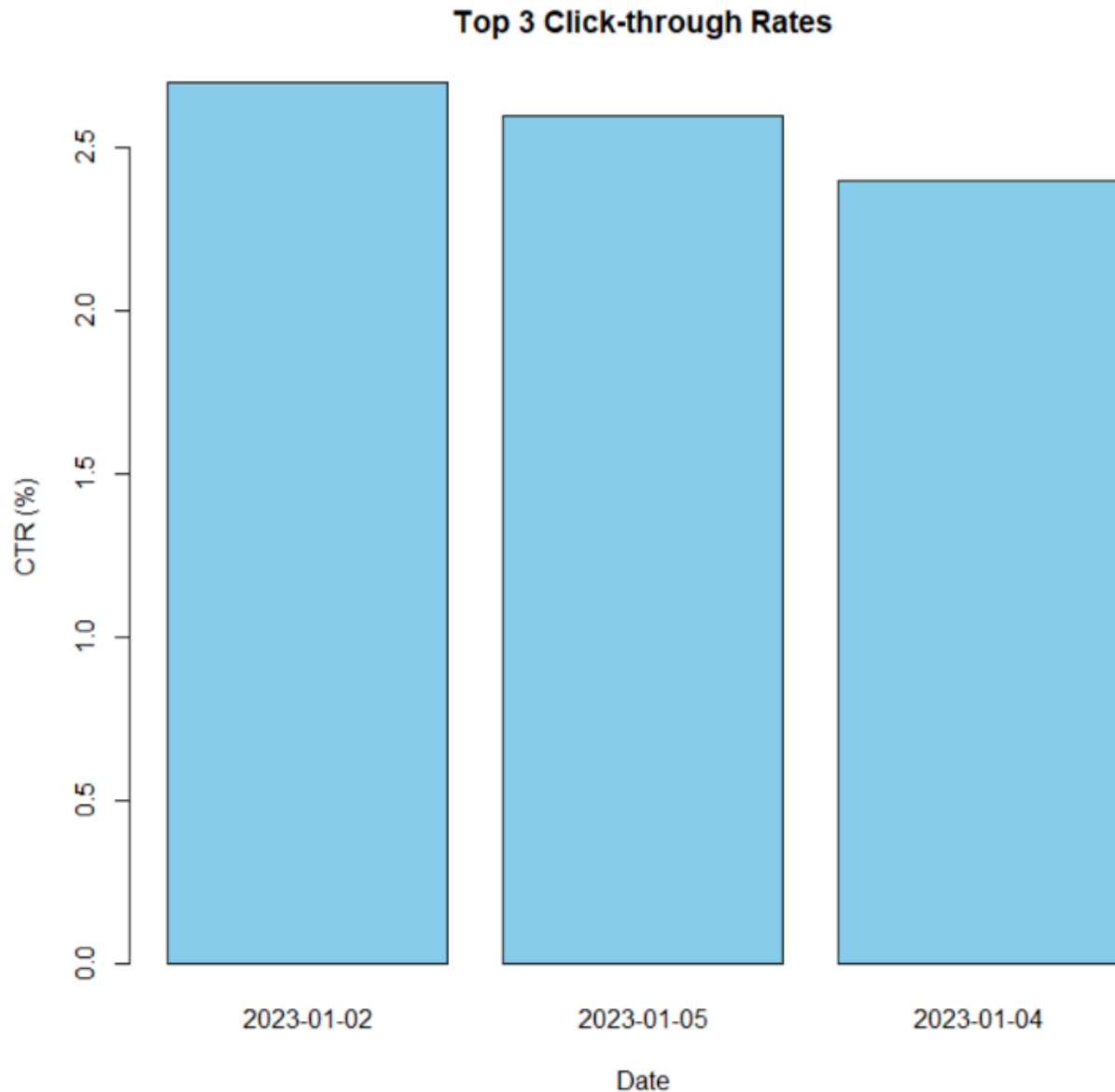
data <- data.frame(date, page_views, ctr)

# Sort by Click-through Rate
top <- data[order(-data$ctr), ]

# Select Top 3 Days
top3 <- top[1:3, ]

# Bar Chart
barplot(top3$ctr,
        names.arg = top3$date,
        col = "skyblue",
        main = "Top 3 Click-through Rates",
        xlab = "Date",
        ylab = "CTR (%)")
```

Output:



4. Tableau – Dashboard (In R Program)

Code:

```
# Dataset
source <- c("Google","Facebook","Instagram","Direct","Twitter")
page_views <- c(1500,1600,1400,1650,1800)
# Display two charts together
par(mfrow = c(1,2))
# Bar Chart
barplot(page_views,
        names.arg = source,
        col = "skyblue",
        main = "Page Views by Source",
```

```
      xlab = "Traffic Source",  
      ylab = "Page Views")  
# Pie Chart (Traffic Source Distribution)  
pie(page_views,  
    labels = source,  
    col = rainbow(5),  
    main = "Traffic Sources")
```

Output:

